SIMLAB — Reflection Questions

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Q1. What is the difference between simulation and optimization?

Simulation is a digital replica of real-world operations — it improves repeatability, cuts costs, and allows safe experimentation. Optimization is the process of searching through those experiments to find the best possible outcome. Simulation models reality; optimization finds the best version of it.

Q2. Why can adding more trucks increase queue time?

Beyond a certain point, more trucks cause congestion and jamming at loading and dumping points. This creates a cascading effect where trucks pile up waiting, and queue times increase across the entire fleet.

Q3. What does a machine-learning model learn from data?

It learns patterns in the data and fits a function that maps inputs to outputs. The goal is for this function to generalize — producing accurate outputs for new, unseen inputs.

Q4. What is a constraint in optimization? Give one mining example.

A constraint is a limit on the problem representing a real-world requirement. In mining, an example is a minimum production target (e.g. at least 10,000 tonnes per shift) or a maximum fleet size (e.g. no more than 15 trucks available).

Q5. What is a Pareto front? Give one example of a mining trade-off.

A Pareto front is the set of optimal trade-off solutions in a multi-objective problem — no solution on the front can improve one objective without worsening another. A mining example is emissions vs. production: pushing output higher means more fuel burn and higher emissions.

Q6. Why is uncertainty important in mine planning and dispatching?

Real mines are unpredictable — equipment fails, ore grades vary, schedules shift. Incorporating uncertainty produces more robust plans that hold up when reality deviates from the model, rather than plans optimized for ideal conditions that rarely exist.

Q7. Which topic was most intuitive? Which was most difficult?

DES and ML were most intuitive due to prior exposure. The trickiest area was deterministic single-objective optimization with PuLP — not conceptually hard, but it wasn't always clear which solver functions were being used internally or why specific solutions were returned.

Q8. How could your background help one part of the SIMLAB project?

I can contribute to the DES and ML components. Knowledge of ML principles helps with model building, while my Mining Engineering background means I can define realistic variables, constraints, and scenarios — grounding the models in actual mine operations rather than abstract setups.